

Program 1 requirements

Program 1 requirements

Write a program to

- calculate the linear regression parameters β_0 and β_1 and correlation coefficients $r_{x,y}$ and r^2 for a set of n pairs of data,
- given an estimate, x_k calculate an improved prediction, y_k where
$$y_k = \beta_0 + \beta_1 x_k$$
- enhance the linked list developed in program 1 to store the n data sets, where each record holds two real numbers

Table 1 contains historical estimated and actual data for 10 programs. For program 11, the developer has estimated a proxy size of 386 LOC.

Thoroughly test the program. At a minimum, run the following four test cases.

- Test 1: Calculate the regression parameters and correlation coefficients between estimated proxy size and actual added and modified size in Table 1. Calculate plan added and modified size given an estimated proxy size of $x_k = 386$.
- Test 2: Calculate the regression parameters and correlation coefficients between estimated proxy size and actual development time in Table 1. Calculate time estimate given an estimated proxy size of $x_k = 386$.
- Test 3: Calculate the regression parameters and correlation coefficients between plan added and modified size and actual added and modified size in Table 1. Calculate plan added and modified size given an estimated proxy size of $x_k = 386$.
- Test 4: Calculate the regression parameters and correlation coefficients between plan added and modified size and actual development time in Table 1. Calculate time estimate given an estimated proxy size of $x_k = 386$.

Expected results are provided in Table 2.

Program Number	Estimated Proxy Size	Plan Added and Modified size	Actual Added and Modified Size	Actual Development Hours
1	130	163	186	15.0
2	650	765	699	69.9
3	99	141	132	6.5
4	150	166	272	22.4
5	128	137	291	28.4
6	302	355	331	65.9
7	95	136	199	19.4
8	945	1206	1890	198.7
9	368	433	788	38.8
10	961	1130	1601	138.2

Table 1

Continued on next page

Program 1 requirements, Continued

Expected results

Test	Expected Values					Actual Values				
	β_0	β_1	$r_{x,y}$	r^2	y_k	β_0	β_1	$r_{x,y}$	r^2	y_k
Test 1	-22.55	1.7279	0.9545	0.9111	644.429					
Test 2	-4.039	0.1681	0.9333	.8711	60.858					
Test 3	-23.92	1.43097	.9631	.9276	528.4294					
Test 4	-4.604	0.140164	.9480	.8988	49.4994					

Table 2

Regression

Overview

Linear regression is a way of optimally fitting a line to a set of data. The linear regression line is the line where the distance from all points to that line is minimized. The equation of a line can be written as

$$y = \beta_0 + \beta_1 x$$

In Figure 1, the best fit regression line has parameters of $\beta_0 = -4.0389$ and $\beta_1 = 0.1681$.

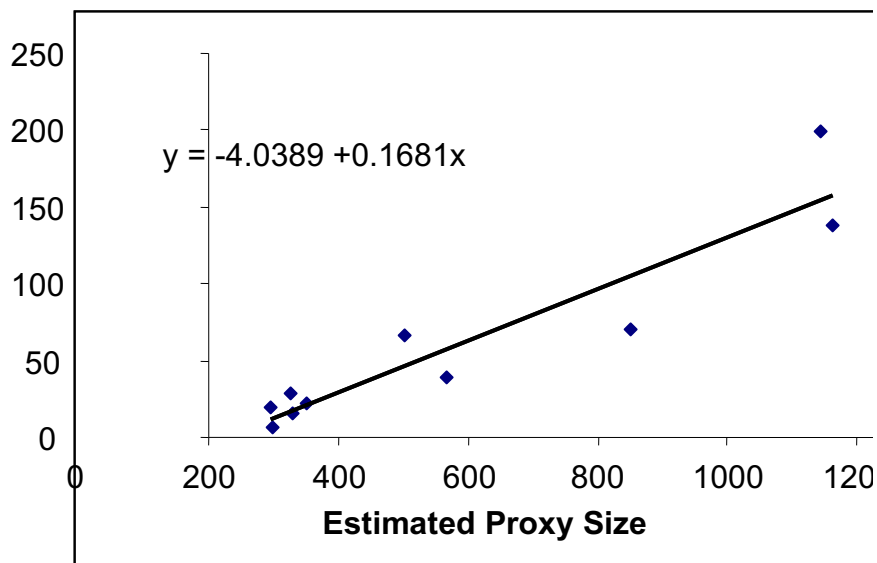


Figure 1

Continued on next page

Regression, Continued

Using regression in the PSP

Looking at Figure 1, how many hours do you think it would take to develop a program with an estimated proxy size of 500?

Using PROBE method A for time, the estimate would be
 $TimeEstimate = \beta_0 + \beta_1(500)$ or an estimate of 80.011 hours.

The PSP PROBE method uses regression parameters to make better predictions of size and time based on your historical data.

PROBE methods A and B differ only in the historical data (x values) used to calculate the regression parameters. In PROBE method A, **estimated proxy** size are used as the x values. In PROBE method B, **plan added and modified** size are used as the x values.

PROBE methods for size and time differ only in the historical data (y values) used to calculate the regression parameters. To predict improved size estimates, **actual added and modified LOC** are used as the y values. To predict time estimates, **actual development times** are used as the y values.

Historical Data Used		x values	y values
Size Estimating	PROBE A	Estimated Proxy Size	Actual Added and Modified Size
	PROBE B	Plan Added and Modified Size	Actual Added and Modified Size
Time Estimating	PROBE A	Estimated Proxy Size	Actual Development Time
	PROBE B	Plan Added and Modified Size	Actual Development Time

Correlation

Overview

The correlation calculation determines the relationship between two sets of numerical data.

The correlation $r_{x,y}$ can range from +1 to -1.

- Results near +1 imply a strong positive relationship; when x increases, so does y .
- Results near -1 imply a strong negative relationship; when x increases, y decreases.
- Results near 0 imply no relationship.

Using correlation in the PSP

Correlation is used in the PSP to judge the quality of the linear relation in various historical process data that are used for planning. For example, the relationships between estimated proxy size and actual time or plan added and modified size and actual time.

For this purpose, we examine the value of the relation r_{xy} squared, or r^2 .

If r^2 is	the relationship is
$.9 \leq r^2$	predictive; use it with high confidence
$.7 \leq r^2 < .9$	strong and can be used for planning
$.5 \leq r^2 < .7$	adequate for planning but use with caution
$r^2 < .5$	not reliable for planning purposes

Limitations of correlation

Correlation doesn't imply cause and effect.

A strong correlation may be coincidental.

From 1840 to 1960, no U.S. president elected in a year ending in 0 survived his presidency.
Coincidence or Correlation?

Many coincidental correlations may be found in historical process data.

To use a correlation, you must understand the cause-and-effect relationship in the process.

Calculating regression and correlation

Calculating regression and correlation

The formulas for calculating the regression parameters β_0 and β_1 are

$$\beta_1 = \frac{\left(\sum_{i=1}^n x_i y_i \right) - (n x_{avg} y_{avg})}{\left(\sum_{i=1}^n x_i^2 \right) - (n x_{avg}^2)}$$

$$\beta_0 = y_{avg} - \beta_1 x_{avg}$$

The formulas for calculating the correlation coefficient $r_{x,y}$ and r^2 are

$$r_{x,y} = \frac{n \left(\sum_{i=1}^n x_i y_i \right) - \left(\sum_{i=1}^n x_i \right) \left(\sum_{i=1}^n y_i \right)}{\sqrt{\left[n \left(\sum_{i=1}^n x_i^2 \right) - \left(\sum_{i=1}^n x_i \right)^2 \right] \left[n \left(\sum_{i=1}^n y_i^2 \right) - \left(\sum_{i=1}^n y_i \right)^2 \right]}}$$

$$r^2 = r * r$$

where

- Σ is the symbol for summation
 - i is an index to the n numbers
 - x and y are the two paired sets of data
 - n is the number of items in each set x and y
 - x_{avg} is the average of the x values
 - y_{avg} is the average of the y values
-

An example

An example

In this example, we will calculate the regression parameters (β_0 and β_1 values) and correlation coefficients $r_{x,y}$ and r^2 of the data in the Table 3.

n	x	y
1	130	186
2	650	699
3	99	132
4	150	272
5	128	291
6	302	331
7	95	199
8	945	1890
9	368	788
10	961	1601

Table 3

$$\beta_1 = \frac{\left(\sum_{i=1}^n x_i y_i \right) - (n x_{avg} y_{avg})}{\left(\sum_{i=1}^n x_i^2 \right) - (n x_{avg}^2)}$$

1. In this example there are 10 items in each dataset and therefore we set $n = 10$.
2. We can now solve the summation items in the formulas.

n	x	y	x^2	$x*y$	y^2
1	130	186	16900	24180	34596
2	650	699	422500	454350	488601
3	99	132	9801	13068	17424
4	150	272	22500	40800	73984
5	128	291	16384	37248	84681
6	302	331	91204	99962	109561
7	95	199	9025	18905	39601
8	945	1890	893025	1786050	3572100
9	368	788	135424	289984	620944
10	961	1601	923521	1538561	2563201
Total	$\sum_{i=1}^{10} x_i = 3828$	$\sum_{i=1}^{10} y_i = 6389$	$\sum_{i=1}^{10} x_i^2 = 2540284$	$\sum_{i=1}^{10} x_i y_i = 4303108$	$\sum_{i=1}^{10} y_i^2 = 7604693$
	$x_{avg} = \frac{3828}{10} = 382.8$	$y_{avg} = \frac{6389}{10} = 638.9$			

Continued on next page

An example, Continued

An example, cont. 3. We can then substitute the values into the formulas

$$\beta_1 = \frac{(4303108) - (10 * 382.8 * 638.9)}{(2540284) - (10 * 382.8^2)}$$

$$\beta_1 = \frac{1857399}{1074926} = 1.727932$$

$$r_{x,y} = \frac{10(4303108) - (3828)(6389)}{\sqrt{[10(2540284) - (3828)^2][10(7604693) - (6389)^2]}}$$

$$r_{x,y} = \frac{18573988}{\sqrt{[10749256][35227609]}} \quad r_{x,y} = \frac{18573988}{19459460.1}$$

$$r_{x,y} = 0.9545$$

$$r^2 = 0.9111$$

4. We can then substitute the values in the β_0 formula

$$\beta_0 = y_{avg} - \beta_1 x_{avg}$$

$$\beta_0 = 638.9 - 1.727932 * 382.8 = -22.5525$$

5. We now find y_k from the formula $y_k = \beta_0 + \beta_1 x_k$

$$y_k = -22.5525 + 1.727932 * 386 = 644.4294$$
